Acta Physiol Plant (2016) 38:280
DOI 10.1007/s11738-016-2295-3

REVIEW

Aconitum biotechnology: recent trends and emerging perspectives

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Received: 1 June 2015 / Revised: 19 October 2016 / Accepted: 31 October 2016 / Published online: 8 November 2016

Abstract The genus *Aconitum* (consists more than 250 species) is one of the most important clades of highly valued medicinal plants. *Aconitum* species are very essential in the traditional device of medication and feature excessive business demand in the herbal marketplace. Some of biologically energetic compounds, e.g., aconitine, indaconitine, pseudoacontine, and so on, had been recognized, and new formulations primarily based on those compounds are being produced as rapid rate. This has led to extensive and rather unregulated exploitation of the species in the wild making the genus a threatened group. Conventional breeding and propagation methods have contributed significantly, but these could not meet up with the ever increasing demands of herbal drug industry globally. Biotechnological interventions, therefore, emerge as an alternative approach in terms of higher production and conservation as well. In recent years, several reports have been published on in vitro propagation of various important *Aconitum* species. However, advanced biotechnological approaches, such as synthetic seed production and hairy root cultures, are still lacking with only a few reports available. The current review presents an updated overview and critical assessment of secondary data concerning the past and recent biotechnological approaches and interventions in genus *Aconitum*. This review also attempts to provide a detailed account of work explored so far in micropropagation and emphasizes over the areas not attempted yet, which will act as a baseline data as well as valuable information for different stakeholders and researchers working on various aspects of *Aconitum* biotechnology.

Keywords *Aconitum* · Micropropagation · Synthetic seed · Hairy root culture

Abbreviations

MS	Murashige and Skoog medium
BAP	6-Benzylaminopurine
NAA	a-Naphthaleneacetic acid
2,4-D	2,4-Dichlorophenoxyacetic acid
Kn	Kinetin
m amsl	Meter above mean sea level
PCR	Polymerase chain reaction
RAPD	Random amplified polymorphic DNA
ISSR	Inter-simple sequence repeat
UPGMA	Unweighted pair-group method with arithmetic mean

Introduction

Steady depletion of biodiversity is a matter of growing concern all over the world. This includes over 21,000 plant species recorded for their medicinal value (WCMC 1992). Among these, the flowering plants constitute an important resource of phytochemical and pharmacological values in

Communicated by A. K. Kononowicz.

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the world of modern medicine (Harnischfeger 2000). The herbal drugs have lesser side effects and relatively lower cost as compared to the synthetic medicine and thus gaining popularity (Gupta et al. 1998). Most of the medicinal plants traded commercially continue to be harvested from the wild, because collecting plant material from wild is considered as most cost effective. Declining population of threatened medicinal plants in the wild, illegal collection, and overexploitation has been hot issues among various concerned departments and workers (Jacobson et al. 1991; Sheldon et al. 1998; Dhar et al. 2000).

As the sources of elite plant material are getting depleted, there is a need for rapid multiplication and conservation of important medicinal plants. The conventional breeding and propagation strategies have contributed appreciably to the growth of pharmaceutical industry over the past several years in a variety of approaches, including the varietal development. However, in view of the limitations of the conventional breeding strategies coupled with the demand for vertical increase in productiveness at less expensive value of manufacturing, the utility of biotechnology-based tools affords attractive opportunity methods. In this regard, plant propagation via tissue culture has been evolved as a valuable method compared to the conventional plant multiplication.

Genus *Aconitum* has been reviewed broadly on the basis of its phytochemical constituents and uses (Gajalakshami et al. 2011; Shyaula 2011; Nyirimigabo et al. 2014) and taxonomy and ecology (Novikoff and Mitka 2011). Many others have reviewed it on the basis of appearance and availability in different parts of the world (Novikoff and Mitka 2011; Shyaula, 2011). Some authors have put focus on only one species, i.e., *A. balfourii* (Sharma and Gaur 2012). Although a number of studies based on micropropagation, synthetic seed production, callus-mediated shoot organogenesis, adventitious shoot regeneration, genetic transformation, etc of genus *Aconitum* have been published (Giri et al. 1993, 1997; Watad et al. 1995; Shiping et al. 1998; Cole and Kuchenreuther 2001; Pandey et al. 2002; Mengliang et al. 2007; Mitka et al. 2007; Bist et al. 2011; Hatwal et al. 2011; Rawat et al. 2013a, b; Zhao et al. 2015), a detailed review covering different aspects of research on genus *Aconitum* has not been attempted yet. Thus, the current review becomes very unique and important for the researcher and readers all across the world working on genus *Aconitum*. In the present study, the cutting-edge evaluation gives an up to date evaluation and critical assessment of published information regarding the current biotechnological techniques and interventions in genus *Aconitum*. Review also attempts to provide a detailed account of research work explored so far in tissue culture and emphasizes the work not attempted yet. The extent of research in terms of explored and remaining biotechnological works in genus *Aconitum* is depicted in Fig. 1. This review also indicates if unexplored areas, such as metabolic engineering, are taken into consideration, might change the face of future research focused on genus *Aconitum*.

Aconitum at a glance

The genus *Aconitum*, normally referred as aconite, monkshood, or wolfsbane, belongs to the Ranunculaceae family of flowering plants. There are over 250 species of *Aconitum* (Lane 2004) widely distributed over the north temperate regions of the world. This species is having major centers of diversity in the mountains of South-East Asia and Central Europe (Kadota 1987). About 28 species have been reported from India, mostly confined to the alpine and sub-alpine regions of Himalaya from Kashmir to Nepal, extending eastwards to Arunachal Pradesh and further to Myanmar, where *Aconitum* were used in local and traditional systems as medicine (Shah 2005). Aconites are herbaceous perennial plant, growing in moisture retentive, however, well draining soils in mountain meadows. Root is best harvested within the autumn as quickly and dried for later use before the plant dies down.

All *Aconitum* plants, mainly cultivated for tubers, contain poisonous alkaloids that can, in sufficient quantity, be deadly. The tuberous roots of several species have long been used as medicine; a number of them are known to contain highly toxic alkaloids (Anonymous 1988; Bhandary and Shrestha 1982, 1986). Outdoor Europe it was extensively used for its medicinal properties. Chinese folk treatments use the roots of *Aconitum* for treating headaches, paralysis (hemiplegia), rheumatism, arthritis, contusions, bruises, broken bones, and overheating of the body. In the context of India, man has used *Aconitum,* since Vedic times for the cure of various ailments (Rigveda, 4500–1600 BC). The Charak Samhita by Agnivesh and Susruta (800–700 BC) also described about this medicinally important herb. Several *Aconitum* species are listed in Red Data Book and are being covered under various conservation programs involving in situ, ex situ/in vitro methods of conservation.

Medicinal and pharmaceutical importance

Genus *Aconitum* is well known for its medicinal property all across the world. Plants of this genus are used as a traditional and Ayurvedic medicine. To reduce the level of toxicity and harness the medicinal benefits of tubers of this species, studies have been done regarding therapeutic dose determination and processing of tubers (Park et al. 1990)

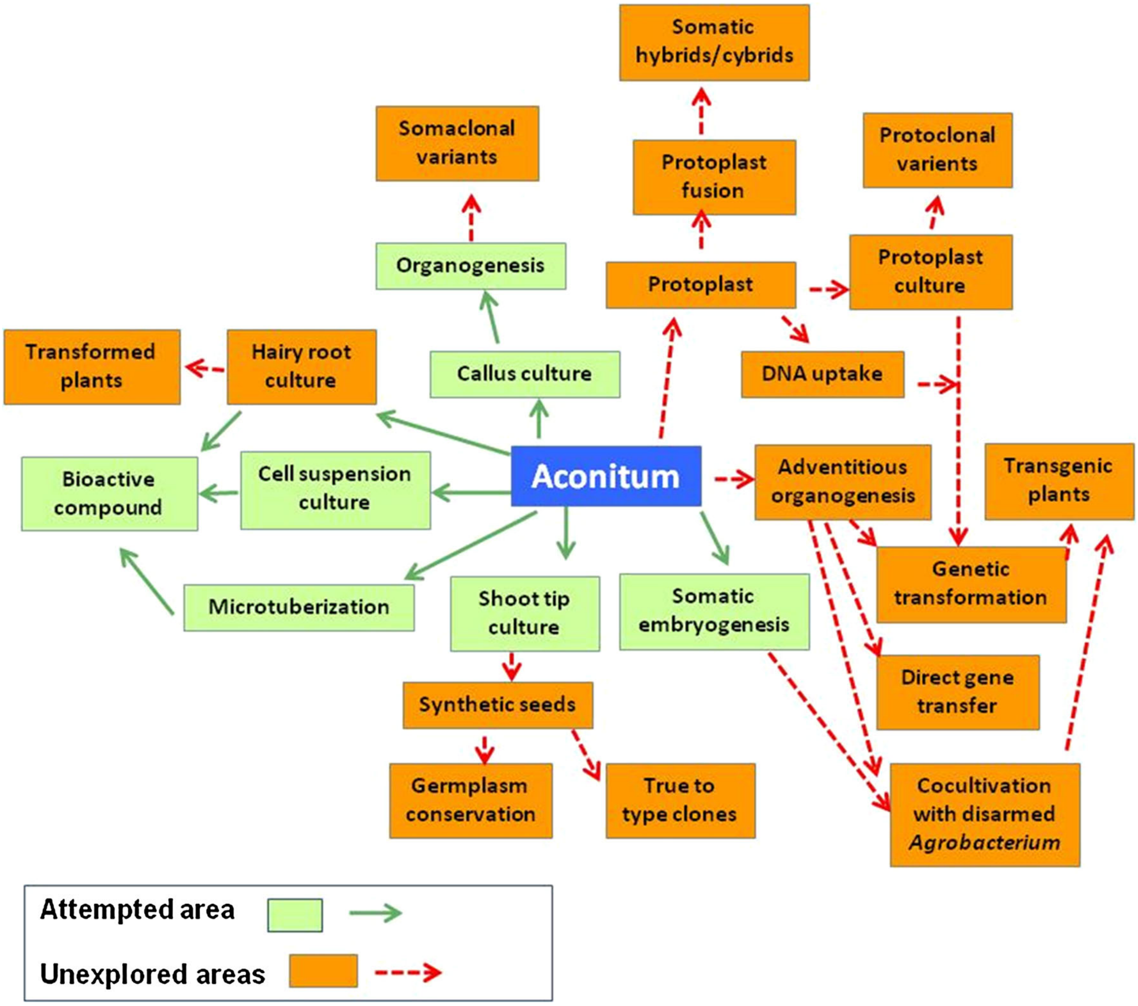

Fig. 1 Tissue-culture-mediated biotechnological intervention in genus *Aconitum*

and multi-drug formulations (Wang et al. 2009). The tuberous roots are used as tonic, aphrodisiac, and in curing high fever (Kirtikar and Basu 1984). Medicinal properties of genus Aconitum are described in Table 1.

Seed germination

Regeneration from seeds is the most customarily used approach for propagation in many plant species (Sharma et al. 2006). Seed germination and early growth of seedlings are important stages in the life cycle of any plant. Seed dormancy along with poor seed availability and lack of superior germplasm (Nautiyal et al. 2009) has been a major limitation in cultivation of most of the *Aconitum* species, e.g., *A. heterophyllum*. Moreover, there are limited seed dormancy studies available in this genus which further limits large-scale propagation. Dormancy plays an important role during maturation and subsequent germination of seed, and it ensures seed survival by prolonging germination when the environment is unsuitable for seedling survival (Fenner 1985). Seed dormancy may be due to the presence of impermeable seed coat, which prevents proper imbibitions, physiological inhibition mechanism of the embryo that prevents radical emergence, or due to the presence of germination inhibitors, such as abscisic acid (Baskin and Baskin 1998). Moreover, factors, such as temperature, light, altitude, mineral, and water availability, also affect seed dormancy.

Seeds of a particular species may possess one or more than one type of dormancy at the time of maturation. Even after the primary dormancy has been overcome, seeds may enter secondary dormancy if any of the environment factors is not suitable for seedling survival. Species, whose seeds germinate to the tune of 50% or more without treating with any of the dormancy breaking process, have been categorized as non-dormant (Baskin and Baskin 1998). Seed dormancy in alpine plants is, generally, considered as an adjustment to the unforgiving environment, where controlled germination can be crucial for seedling

Table 1 Medicinal uses and active ingredients of different species of genus *Aconitum*

Species name	Medicinal use	Active ingredient	References
Aconitum anthora	–	Flavonol glycosides and flavonoids	Mariani et al. (2008)
A. balforii	Rheumatism, cholera, fever	Alkaloids, 9-hydroxysenbusine a, balfourine, 8-*O*-methyl-veratroylpseudaconine	Anonymous (1988); Khetwal and Pande (2004); Khetwal et al. (1992)
A. bisma	Gastrointestinal disorders, joint aches, rheumatism	Palmatisine, vakatidine, vakatisinine	Singh and Singh (1972); Jiang and Pelletier (1988); Jiang and Pelletier (1991)
		Vakognavine	
		6-Acetylheteratisine, 15-deacetylvakgnavine, palmasine, palmadine	
		6,8-Diacetylheteratisine	
		Vakatisine	
		Vakhmadine, vakhmatine	
A. brachypodum	Anti-rheumatic and analgesic properties	–	Singhuber et al. (2009)
A. bulleyanum	Influenza, rashes and snake bite	–	Wangchuk et al. (2010)
A.burnatii	–	Flavonol glycosides	Vitalini et al. (2010)
A. carmechaeli	Analgesic, anti-inflammatory	Sonorine, vneoline, *N*-dethylneoline, 8-*O*-cinnamoylneoline	Csupor et al. (2009); Hikino et al. (1983); Chen et al. (2008)
		Diterpenoid alkaloids, such as aconitine, mesaconitine, and hypaconitine	
A. chasmanthum	Anti microbial	Essential oil	Anwar et al. (2003); Zhang et al. (2009)
A. coreanum	Neurological, antagonistic activity, myorelaxant	Deterpene alkaloids	Ameri (1998); Ameri and Simmet (1999); Dzhakhangirov and Bessonova (2002)
A. deinorrhizon	Rheumatism, rheumatic fever, headache	Aconites	Pullaiah (2006)
A. delavayi	Rheumatism, traumatic injuries blood stasis, swelling,pain, matemesis, hemoptysis, hematochezia, piles hemorrhage	C18-norditerpenoid alkaloids, such as delavaconitine F 1 and Delavaconitine	Jiang et al. (2007)
		G 2	
A. episcopale	–	Diterpenoid alkaloids	Yang et al. (1999)
A. ferox	Stimulant, fever, cephalagia, throat dyspepsia, urinary trouble	14-*O*-Acetylsenbusin A, Bikhaconite, Pseudaconine	Hanuman and Katz (1994a); Hanuman and Katz (1994b); Achmatowiez and Marim (1964); Purushothman and Chandrasekharan (1974); Hanuman and Katz (1993a); Hanuman and Katz (1993b)
		14-*O*-Benzoyl-Lipopseudaconitine, lipobikhaconitine, lipopseudaconitine chasmaconitine	
		Diacetylpseudaconitine, veratroylpseudaconine	
		3,4-Dihydro-6-hydroxy-2(1H)-quinoline	
		Pseudaconitine	
		Veratroylbikhaconine	
A. finetianum	Enteritis, poisonous snake-bites and fractures	Diterpenoid alkaloids, such as Anthranoylly coctonine (inuline) and Lycoctonine	Wu et al. (1996)
A. franchetti	–	Franchetine Norditerpenoid compound	Wang et al. (1997)

Table 1 continued

Species name	Medicinal use	Active ingredient	References
A. gammiei	Stomachic	16-Acetoxycardiopetaline	Liu and Katz (1994); Feunte et al. (1989); Gonzlez et al. (1983); Liu and Katz (1996); Feunte et al. (1989); Arlandini et al. (1987); Feunte et al. (1988); Kopp (1973); Hikino et al. (1983, 1984); Desui et al. (1998); Lu et al. (2004); Chu et al. (1964)
		15-Acetyl-13-dehydrocardiopetamine	
		15-Acetylcardiopetamine Brachyaconitine	
		Cardiopetamine	
		N-Deethylaconitine	
		1,14-Diacetylneoline, 12-Epiacetyldehydronapelline	
		Ephedrine	
		Hokbusine A	
		Senbusine A	
		Lpaconitine	
		Merckonine	
		Myriophyllosides F	
		Napelline	
		Neoline	
		Norepinephrine	
		Taurenine	
A. hemsleyanium	–	Franchetine-type and aconitine -type C19-diterpenoid alkaloid	Tang et al. (2007)
A. heterophylloides		Heterophyllodine	Pelletier et al. (1981)
A. heterophyllum	Anti diarrhoeal	Alkaloids, anthorine	Singh and Chaturvedi (1982); Nisar et al. (2009); Lawson and James (1937); Pelletier et al. (1968, 1982); Aneja et al. (1973); Gonzlez-Coloma et al. (2004); Pelletier et al. (1968, 1978); Pelletier and Parthasarathy (1965); Jacobs and Craig (1942); Ulubelen et al. (2002); Ross et al. (1992); Shaheen et al. (2005)
	Enzyme inhibitor	Anthorine	
		Atidine	
		Atisenol	
		6-Benzoylheteratisine	
		11,13:11,16-Diepoxy-16,17-dihydro-11,12-secohetisan-2-ol	
		Dihydroatisine	
		Heterophyllidine, Heterophylline, hetidine, Hetisinone, *O*-methyl-	
		heterophylline	
		Isoatisine	
		Hetisine	
		N-Deethyl-*N*-formyllyaconitine *O*-Methylaconitine	
		Methyl-*N*-succinoylanthranilate	
A. karacolicum	Anti proliferative	Alkaloids	Chodoeva et al. (2005), Hazawa et al. (2009)
	Anti tumerous		

Table 1 continued

Species name	Medicinal use	Active ingredient	References
A. kirinense	Rheumatic arthritis rheumatoid disease	Diterpenoid alkaloids, such as Kirinines B and C	Feng et al. (1998)
A. kusnezoffii	Analgesic and anti-rheumatic herbal medicine	Water-insoluble glucan (AKP) compound	Zhao et al. (2009); Gao et al. (2010)
	Treat heart failure congestion, neuralgia, rheumatism, gout, etc homeopaths		
A. leave	Antioxidant, anti-inflammatory	Swatinine	Shaheen et al. (2005); Mariani et al. (2008)
		Benzene derivative, alkaloids	
A. napellus	Homeopathic preparations	Flavonol glycosides	Singhuber et al. (2009); Fico et al. (2001); Luis et al. (2006)
		Quercetin3-*O*-(6-transcaffeoyl)-β-glucopyranosyl-(1 → 2)-β-glucopyranosyl-7-*O*-α-rhamnopyranoside and quercetin-3-sophoroside-7-rhamnopyranoside	
A. naviculare	Cold, fever, headache, sedative, analgesic, febrifuge	Navirine	Gao et al. (2004); Shrestha et al. (2006)
		Flavonol glycosides	
A. nasutum	–	Norditerpenoid alkaloid 3-hydroxy talatisamine	Mericli et al. (1996)
A. orientale	–	Diterpenoid alkaloids	Ulubelen et al. (1996)
A. orochryseum	Fever, rheumatism,	Atisiniumchloride	Wangchuk et al. (2007)
A. racemulosum	–	Norditerpenoid alkaloids	Wang et al. (2000)
		Diterpenoid alkaloids	
A. sinomontanum	–	Lappaconitine, 3 rranaconitine, *N*-deacetyllappaconitine and *N*-deacetylranaconitine	Yang and Ito (2002)
A. soongaricum	Psychostimulant, antidepressant, respiratory problems	Acetylsongorine	Zhamerashvili et al. (1981); Pelletier et al. (1979)
		Songoramine	
		Songorine	
		Songorinine	
		12-Acetyl-12-epinapelline	
		Aconine	
A. spicatum	Rheumatism, fever	Bikhaconitine	Dunstan and Andrews (1905); Gao et al. (2005, 2006)
		Spicatine A, spikatine B	
A. sungpanense	Rheumatism arthritis neurological disorder and as an analgesic medicine	Trans-2,2V,4,4Vtetramethyl-6,6V-dinitroazobenzene	Wanga et al. (2004)
A. tanguticum	–	6-Benzoylheteratisine	Ameri and Simmet (1999)
A. taipeicum	Anti-inflammatory and analgesic	Amide alkaloids	Xu et al. (2010)
A. transsectum	–	Norditerpenoid alkaloids	
A. variegatum	–	Norditerpene and diterpene alkaloids and sachaconitine, karakoline, talatizamine hydroxytalatizamine etc	Díaz et al. (2005)
A. violaceum	Renal pain, high fever	Diterpenoid alkaloids	Chauhan (1999); Miana et al. (1971)
		Indaconitine	
A. vulparia	Rheumatism, neuralgia and chronic skin disorders	Flavonol glycosides	Fico et al. (2003)

survival (Bliss 1971; Kaye 1997). It seems that seed dormancy acts a critical role in alpine plant community composition for the reason that germination rate and timing does affect seedling survival and competitive functionality; this in the end affects populations, successional, and community patterns of high elevation plants (Harper 1977; Kaye 1997).

The present review discusses some of the studies that have been conducted towards improved seed germination and overcoming the seed dormancy barriers in genus *Aconitum*. Dalteskaya (1985) found that it is the presence and type of GA that plays an important role in breaking seed dormancy and there exists variation among different species of the same genus towards dormancy breaking treatments. In *A. balfourii,* Stapf GA_3 at a concentration of 250 μM significantly enhanced germination of seeds to 42.5% as compared to control (27.5%) in 15 weeks, while in *A. heterophyllum* Wall, a higher concentration of BAP (250 μM) augments germination up to 42.5% as compared to 25% in controls (Pandey et al. 2000). In the same study, it was also observed that thiourea and potassium nitrate have a promoting effect on seed germination in *A. balfourii,* but fail to induce germination in *A. heterophyllum*. Thus, nitrogen containing compounds also seem to play significant role in breaking seed dormancy.

Seed germination of *A. heterophyllum* has been improved with the treatment of BAP (Pandey et al. 2000). It has been suggested that cytokinins may play a 'permissive' role by decreasing the level of germination inhibitors and making the seed more responsive to gibberellins (Khan 1971; Walker et al. 1989). Later, Pandey et al. (2005) reported that a hot water treatment at 40–50 °C for 90 s significantly enhances germination in seeds of *A. heterophyllum*. According to Beigh et al. (2005), chilling treatment could improve germination of *A. heterophyllum* seeds at lower altitude, and thus, lower altitude cultivation can be an approach towards ex situ conservation. Similar findings have been reported in *A. sinomontanum* by Dosmann (2002).

Temperature is one of the most important factor which affects seed germination in *Aconitum* species. In *A. lycoctonum,* Vandelook et al. (2009) reported growth and germination of embryos to occur only at low temperatures of about <10 °C. Pre-treatments, such as high temperature and $GA_{3,}$ could not overcome the chilling requirement. Natural as well as artificially induced seed germination and establishment of seedlings is found to be rare in *A. atrox* (Bhadula et al. 2000; Nautiyal et al. 1985). Rawat et al. (1992), keeping the lengthy (5–7 years) cultivation cycle in mind, successfully attempted vegetative propagation of *A. atrox* under alpine conditions (3600 masl) with an aim to provide a constant supply of raw materials and achieve species conservation too. Similarly, Kuniyal et al. (2003) conducted growth and propagation experiments at lower altitudes (1900 masl) in which tuber segments were treated with GA_3, IBA, and kinetin either singly or in combination for 48 h at room temperature. The results showed no significant difference when compared to propagation at higher altitudes.

On the basis of the above-mentioned studies and an account of commercial relevance studies on dormancy and germination pattern in important alpine plants would be pertinent from the cultivation point of view. Moreover, the effects of externally applied plant growth substances in synchronizing and improving germination in alpine plants are limited. Further research still needs to be done to modify and standardize effective cultivation strategies that would have economic benefits to the farmers and also the consumers.

Micropropagation

Micropropagation, defined as in vitro regeneration of plants from cells, tissues, organs, etc (Vasil and Vasil 1980), offers the possibility of multiplying plants rapidly under controlled and protected conditions. Plant tissue culture techniques have been widely used for the commercial propagation of a number of economically important species (Bhojwani and Razdan 1996; Ravishankar and Venkataraman 1997). This technique is useful not only for rapid and mass production of existing stock of plant species, but also offers advantages for the conservation of important and threatened taxa. It also provides novel approaches for the induction of genetic variability. In addition to various aspects related to plant biotechnology, such as in vitro propagation, biosynthesis and biotransformation and production of secondary metabolites, storage of plant cells, and organs etc, genetic engineering of higher plant can only be handled through the use of in vitro techniques (Bajaj 1989). The promising biotechnological interventions which have been attempted in *Aconitum* through micropropagation/tissue culture are presented in Fig. 1.

Several studies have been reported on the micropropagation of *Aconitum* (Watad et al. 1995; Cervelli 1987; Hatano et al. 1988; Shipping et al. 1988; Giri et al. 1993). The first study on tissue culture of *Aconitum* species was reported by Cervelli (1987). Later, Hatano et al. (1988) did successful clonal multiplication through shoot tip-tissue culture of *A. carmechaeli*, which is an important medicinal species of China used in rheumatism and neuralgia (Shoyama et al. 1993). Shiping et al. (1998) reported that microtubering in *A. carmechaeli* Debx was affected by temperature conditions with 15 °C found to be best suited for in vitro microtubering done under dark conditions for 6 weeks. It was also observed that at 20 °C, higher levels of mesaconitine and hypaconitine alkaloids were reported as compared to that at 15 and 10 °C.

In vitro regeneration in *A. heterophyllum* has been reported by Giri et al. (1993). Callus formation was observed on MS medium supplemented either with 1 mg/L 2,4-D and 0.5 mg/L kinetin along with 10% coconut water or with 0.5 mg/L NAA and 1.0 mg/L BAP while maintaining the cultures on 1 mg/L NAA containing MS medium. In another study carried out by Jabeen et al. (2006), callus was reported to be induced in MS medium containing rather low levels of NAA and BAP, i.e., 0.5 and 0.25 mg/L, respectively. Embryos cultured in ¼ strength MS medium supplemented with 1 mg/L IBA gave rise to complete plantlets after 4 weeks of culture (Giri et al. 1993). Enhanced multiplication rate has been achieved in *A. napellus* when cultured on floating membrane rafts in liquid medium (Watad et al. 1995). The method appeared to significantly reduce the requirements for different growth components when compared to conditions of solidified agar medium. Chandra (2003) reported in vitro shoot regeneration and multiplication of *A. violaceum* through callus formation using shoot tip explants. MS medium without any plant growth substances was found suitable for multiplication of shoots, and spontaneous root induction was also observed (Chandra 2003). Srivastava et al. (2010) have standardized the protocol for sterilization of seeds and nodal segments for conservation purposes of this important medicinal plant.

Leaf segments from in vitro grown axillary buds were used to induce callus formation in *A. balfourii* when cultured on MS medium supplemented with BAP and NAA at concentration of 4.5 and 26.9 µM, respectively (Pandey et al. 2004). Shoots were induced at similar concentration of BAP and a low concentration (1.1 µM) of NAA followed by multiplication of shoots at 1.1 µM BAP and in vitro rooting in 12.3 µM IBA containing MS medium. In the same species, a comparative analysis of in vitro tuber-raised and seed-raised plants of comparable age with respect to chromosome, protein profile, and alkaloid content revealed an identical profile in all the cases (Pandey et al. 2004). Relative similarity was also observed in development of callus-originated somatic embryos and sexual embryos in situ in *A. balfourii* (Batygina 2004). Rawat et al. (2013a, b) developed protocol for in vitro propagation in *A. violaceum* and studied effects of plant growth regulators on indirect shoot organogenesis. Tissue culture studies based on micropropagation of the genus *Aconitum* is depicted (Table 2).

Synthetic seed production and assessment of genetic fidelity

Varietal improvement through in vitro propagation techniques has significantly enhanced the development of pharmaceutical sector during the last several decades. In vitro techniques which allow rapid and mass propagation have emerged as an alternative towards 'recovery' of endangered species and subsequently reduce the risk of extinction (Nadeem et al. 2001).

Moreover, in vitro short- to long-term storage of plant species in the form of slow growing and cryopreserved cultures is also a practice greatly followed by various plant conservation programs. Alginate encapsulation of explants in the form of synthetic seeds is one of the many in vitro techniques utilized towards germplasm storage and conservation (Srivastava et al. 2009; Mishra et al. 2011; Rawat et al. 2013b). The alginate-encapsulation method maintains

Table 2 Detail account of work carried out on in vitro propagation of genus *Aconitum* in the last two decades

Species name	Explants used	Medium used	Results	References
A. balforii	Leaf and lateral buds, axillury buds, leaf	MS + 4.5 µM BAP + 26.9 µM NAA, 4.5 µM BAP + 1.1 µM NAA	Globular embryo, organogenesis, callus, plantlets, rooting	Singh et al. (1998); Pandey et al. (2002); Bist et al. (2011)
A. carmichaeli	Anther, shoot tip	–	Embryo, multiple shoots, plantlets	Hatano et al. (1988); Shiping et al. (1998)
A. heterophyllum	Leaf, petiole, embryogenic and non-embryogenic callus	MS + 2, 4 D 1 mg/L +Kn 0.5 mg/L, NAA 0.5 mg/L + BAP 0.25 mg/L	Embryo, plantlets, transgenic hairy roots	Giri et al. (1993, 1997)
A. napellus	Shoot tip, stem nodes	–	Plantlets, rosette plantlets	Watad et al. (1995); Cervelli (1987)
A. noveboracense	Stem nodes, leaf	–	Plantlets, callus, embryoids, shoots	Cervelli (1987)
A. uncinatum	Shoot apex	–	Multiple shoots, plantlets	Lim and Kitto (1995a, b)
A. violaceum	Shoot tip, callus	MS + 0.1 µM BAP + 0.5 µM NAA	Multiple shoots, plantlets	Chandra (2003); Rawat et al. (2013a, b)

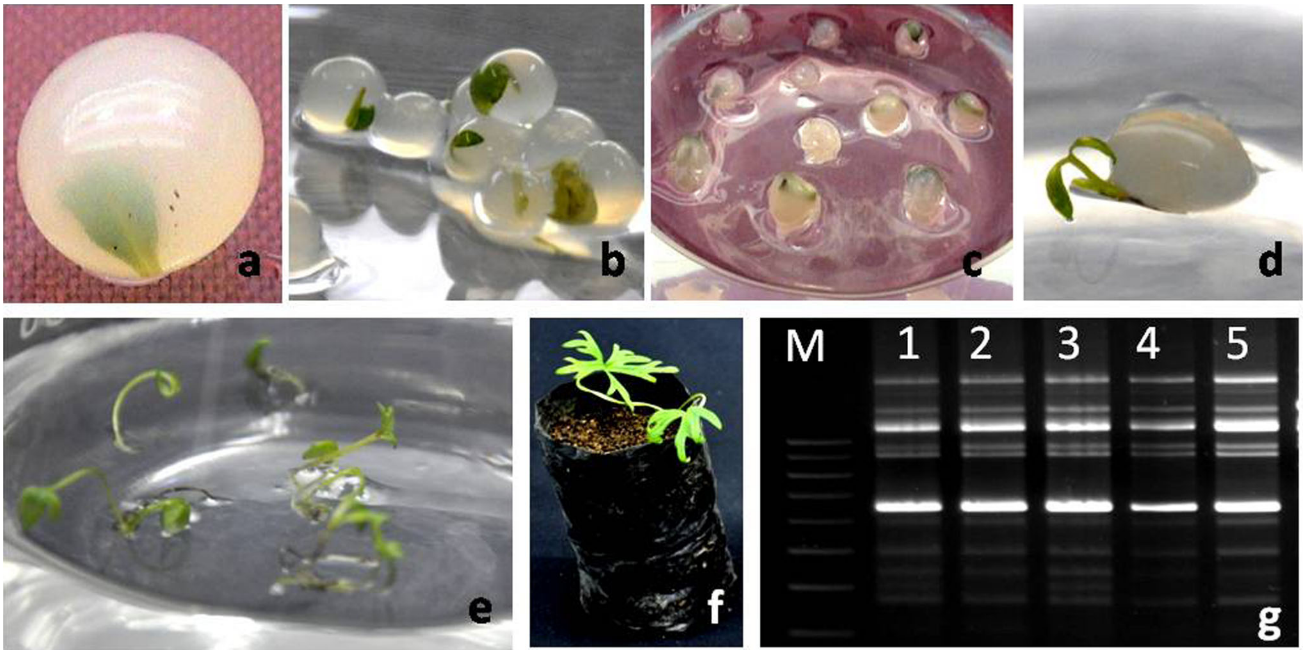

Fig. 2 Synthetic seed production in *A. violaceum*; **a**, **b** synthetic seeds of *Aconitum violaceum*; **c** synthetic seeds on MS medium; **d** shoot induction; **e** multiplication; **f** hardened plant; and **g** genetic fidelity analysis using molecular markers (*M* molecular weight marker, *1* control, and *2–5* synthetic seed-derived plants)

the viability and productivity of cultures and, therefore, provides the benefits of clonal multiplication as well as seed storage and propagation (Ikhlaq et al. 2010; Rawat et al. 2013b). The encapsulated plant tissue is provided with a nutrition rich micro-environment and also gets protected from the environmental stresses during short- to long-term storages. When compared to cryopreservation, the alginate-encapsulation-based short-term storage is cost effective, simpler, and easier in handling (Germana et al. 2011; Rawat et al. 2013b).

Synthetic seed production in *A. violaceum* has been reported by Rawat et al. (2013b) for the first time. Alginate encapsulation of in vitro grown microshoots was carried out with a re-growth rate of approximately 85.43%. Following re-growth on plant growth regulator-free MS medium, multiple shoots were formed in 39.86% of encapsulated microshoots and healthy root development was observed in all the cultures following 2 weeks of transfer. A survival rate of 87% was obtained when plants were transferred to pots with a mixture of soil, sand, and compost (1:1:1 v/v) and finally shifted to green house (Fig. 2).

The most important objective of tissue culture is to achieve increased plant yield without compromising quality, and the emergence of genetic variations due to in vitro culture conditions is one of the major challenges faced during in vitro propagation of plants. Such in vitro induced variations may be a result of stress caused due to the long-term maintenance of plant tissue cultures under artificial in vitro conditions, use of higher concentrations of plant growth regulators, and other growth-promoting substances and regular sub-culturing of cells and tissues (Rani and Raina 2000). Thus, for mass-scale production, efficient, reliable and cost-effective propagation methods along with long-term genetic stability of plantations are of pivotal importance.

The genetic stability of plants recovered following cryogenic storage has been evaluated in a broad range of plant systems (Aronen et al. 1999; Hirai and Sakai 2000; Bekheet et al. 2007). The genetic fidelity of *A. violaceum* plants, developed after the storage of encapsulated microshoots, was assessed through RAPD and ISSR analysis (Rawat et al. 2013b). Genetic similarity of tissue culture-raised (TR) and synthetic seed-derived (SR) plants was established through RAPD and ISSR-based cluster analysis which revealed an average similarity coefficient of 0.966 and 0.974, respectively. Results of the study further support the practicability of this cost effective and easy germplasm conservation approach for storage and regeneration of true-to-type plants.

Hairy root culture/genetic transformation

Plant roots have the ability to produce specific secondary metabolites, which are used for pharmaceutical purposes. Due to increased demand, in the recent times, of plant-based drugs in the indigenous as well as international market, a large number of plant species are in high demand as raw materials for various pharmaceutical firms. Medicinal properties as diverse as antipyretic, antihelminthic, expectorant, antimicrobial, anticancer, and hepatoprotective, etc have been ascribed to different secondary metabolites produced by plants.

Hairy root cultures obtained through transformation of the plant tissue with a gram-negative soil bacterium, *Agrobacterium rhizogenes*, have emerged as a potential means for in vitro production of secondary metabolites. These hairy roots are unique in their genetic and biosynthetic stability, and their fast growing nature compared to untransformed root cultures offers an additional benefit. Hairy root cultures are an exceptional useful system for metabolic studies, not only for production of valuable natural plant products, but also for studies of plant genetic transformation of organic contaminants (Morgan et al. 2000). These prompt growing hairy roots can be used as an alternative and continuous source for the production of valuable secondary metabolites or active ingredients. Hairy root cultures have been established for various economically important plant species, and secondary metabolite production in such cultures has been evaluated in *Artemisis annua* (Weathers et al. 1994), *Catharanthus roseus* (Morgan et al. 2000), *Physia minima* (Azlan et al. 2002) potato (Dobigny et al. 1995), and some cultivars of tomato (Peres et al. 2001). There are limited studies on hairy root induction in alpine plant species of commercial importance (Giri et al. 1997).

Hairy roots have the potential of continued growth on plant growth substance-free medium that can be harvested for continued production of compounds, such as anthraquinone, flavonoids, alkaloids, etc under in vitro conditions (Chang et al. 1998; Li et al. 2000; Sheludko et al. 2002). Furmanowa and Syklowska-Baranek (2000) reported induction of hairy roots in seedlings of *Taxus media* var. Hicksii using LBA 9402 strain of *Agrobacterium rhizogenes*. Following exposure to 100 μM methyl jasmonate, taxol levels in the hairy roots increased from 69 to 210 μg/g dry weights; however, the content of 10-deacylbaccatin III was not affected by the elicitor. Hairy root culture has also been established in *Panax ginseng* (Korean ginseng) and scaled up in bioreactors (Jeong et al. 2002a, b). Transformed hairy roots have been shown to produce elevated levels of active ingredients in several medicinal plant species. For example, Baiza et al. (1998) have reported that the production of tropane alkaloid in hairy root cultures of *Datura stramonium* was much higher than in an untransformed line. The development of hairy roots and optimization of conditions for the production of active ingredients may also help in reducing production cost of such compounds. There are several published reports on hairy root induction in many plant species (Giri et al. 1997; Choi et al. 2004; Pavlov et al. 2005; Mishra et al. 2011), but there is only one study has been reported on hairy root induction in *A. heterophyllum* (Giri et al. 1997). For the first time, hairy root formation in *A. heterophyllum* transformed with *Agrobacterium rhizogenes* was standardized by Giri et al. (1997), and it was found that total alkaloids (aconites) in transformed roots was 3.75% higher than in non-transformed ones.

Plant regeneration from transformed hairy roots could also serve in obtaining plants with higher yield of active principles for commercial applications. For example, the leaves of *Vinca minor* accumulate vincamine, which is used as cerebral vasodilator. Transgenic plants regenerated from hairy roots were able to accumulate higher levels of vincamine and the plants maintained elevated levels of vincamine (Tanaka et al. 1995). Choi et al. (2004) have reported plant regeneration from transformed hairy root cultures of *Catharanthus roseus*. They used hypocotyls explants for hairy root induction and observed up to 80% frequency of adventitious shoot formation. Saxena et al. (2007) have reported improved essential oil (10.3% geraniol) quality in Rose-scented geranium by *Agrobacterium rhizogenes*-mediated Ri-insertion. Plant regeneration from hairy roots of apple root stock has also been reported (Yamashita et al. 2004); the plants developed abundant active roots, compared with the non-transformed control which is useful for modification in morphology of the apple root stock. Similarly, development of transgenic plants with the potential of increased biosynthesis/accumulation of active compounds would help in reducing the pressure on the natural populations.

Molecular studies

A great deal of diversity has been observed in plants and medicinal plants are no exception; this diversity is associated with the genetic makeup of the plant. Molecular marker based analyses are being applied to characterize genetic diversity within species and amongst different species; this is important for proper documentation of biodiversity and proper fingerprinting of different clones. Use of random amplified polymorphic DNA (RAPD), inter-simple sequence repeats (ISSR), restriction fragment length polymorphism (RFLP), amplified fragment length polymorphism (AFLP), and PCR–RFLP based markers is being made for this purpose; some examples in respect of selected medicinal plants of IHR have also appeared (Sharma et al. 2000; Anonymous 2000).

Biochemical markers

Identification has traditionally been based on morphological descriptors, such as plant height and shape, leaf shape,

Table 3 Past evidences of genetic analysis of genus *Aconitum*

Species Name	Work done/marker used	References
A. balforii	RAPD analysis	Hatwal et al. (2011)
A. carmichaeli	RAPD and ISSR analysis	Mengliang et al., (2007); Zhao et al. (2015)
A. columbianum	Isozyme and RAPD analysis	Cole and Kuchenreuther (2001)
A. fimum	ISSR and RAPD analysis	Mitka et al. (2007)
A. lycoctorum	Enzyme electrophoresis	Utelli et al. (2000)
A. noveboracense	Isozyme and RAPD analysis	Cole and Kuchenreuther (2001)
A. plicatum	ISSR and RAPD analysis	Mitka et al. (2007)
A. kongboense	AFLP analysis	Meng et al. (2014)
A. kongboense	ISSR analysis	Meng et al. (2015)

young leaf type, fruit shape, etc. Biochemical markers were also used for characterization of several plant species. Isozymes provide potentially powerful and reliable tools in resolving genetic relatedness/divergence and have been used as molecular markers in genetic, phylogenetic, taxonomic, and evolutionary studies. However, most morphological, physiological, and biochemical traits are influenced by the environment, plant age, and phenology, and thus, with the progression of newer techniques of molecular biology, such efforts were shifted to DNA-based markers.

Molecular markers

Although various types of molecular markers have been used in genetic diversity analysis of various plant species, but in case of *Aconitum,* very limited studies have been reported. Workers have been used RAPD and ISSR markers to assess genetic diversity (Table 3). Since discovered, random amplified polymorphic DNA (RAPD) assay is being used for a number of areas in plant taxonomy (Williams et al. 1990). Presently, it is the most favored DNA markers due to better speed, easy-to-perform, low cost and non-requirement of radioactive materials, etc. Genetic profiling of some populations of *Aconitum fimum*, *A. plicatum*, *A. carmichaeli,* and *A. balfourii* has been reported recently (Table 3). Inter-simple sequence repeat (ISSR) has also been used for genetic diversity studies due to its greater repeatability and stability of map position in the genome comparing genotypes of closely related individual (Zietkiewicz et al. 1994).

Conclusions and future prospects

The value and importance of medicinal plants and the traditional knowledge is being increasingly realized throughout the world. It is estimated that about 75% of biologically active molecules derived from plants and presently in use have resulted from follow-up research to verify the authenticity of folk and ethnomedicinal practices (Farnsworth et al. 1985). As a consequence of increasing demand for medicinal plants by the pharmaceutical industry, there has been a large scale and uncontrolled collection from the wilds, and in the absence of organized cultivation, many of medicinal plants are now placed in the list of critically rare, vulnerable, endangered, or threatened species. Genus *Aconitum* is not untouched from threats, and many species of this genus have already been indexed under the threatened or endangered category (Chaudhary and Rao 1998; CAMP 2003; Bisht and Badoni 2009). Cultivation approaches adopted for conservation and production of *Aconitum* are neither modern nor worthwhile because of the lack of interest and knowledge of the farmers as well as stakeholders, encouraging authority rules, availability of guaranteed markets, beneficial price, and assesses to smooth and suitable agro-techniques. In view of the ecological status and the role of these plants in the economy of the particular region, as well as the importance of the secondary metabolites produced by these plants, it is important to know how best to multiply the species either by the conventional or in vitro technique.

Proficient protocol for in vitro regeneration of *A. balfourii* and *A. heterophyllum* has been reported by several workers (Bist et al. 2011; Pandey et al. 2004; Jabeen et al. 2006). As far as the methods for conservation of endangered and threatened species of medicinal plants are concerned, somatic embryogenesis has been experimented in *A. heterophyllum* (Giri et al. 1993) and proven authentic. Though lot of laboratory work has been done on micropropagation of *Aconitum* (Watad et al. 1995; Cervelli 1987; Hatano et al. 1988; Shiping et al. 1998; Giri et al. 1993; Rawat et al. 2013a; Mahajan et al. 2015), yet challenge remains on commercial application of it (Fig. 1). There has been some progress on large-scale cultivation; however, production is not adequate to meet the requirement of pharmaceutical industries. Hairy root culture, although the success rate are poor in *Aconitum* (Ranunculace), there is a large scope for secondary metabolite and transgenic plant production to fulfill the requirement of the industries. As mentioned earlier, hairy roots have been induced in *A.*

heterophyllum by Giri et al. (1997). Recently, few studied have reported on culture conditions for hairy root induction in *Aconitum coreanus* (Yu et al. 2013; Liu et al. 2015). Moreover, in vitro techniques have been used for plant multiplication, and use of callus and genetically transformed hairy roots for the production of active ingredients is gaining importance.

Protein and isozyme markers have been widely used in the past for identification of plants. Until date, a very slow progress has been made regarding the development of biochemical and genetic markers in *Aconitum* species. Pathak et al. (2011) have reported biochemical variability in the population of *A. balfourii* by soluble protein and isozyme. Recently, Kakiuchi et al. (2015) have reported the diversity in alkaloid profile of *Aconitum* plants. In molecular markers, several attempts have been made to assess the genetic diversity using RAPD and ISSR markers mainly (Cole and Kuchenreuther 2001; Mengliang et al. 2007; Mitka et al. 2007; Zhao et al. 2015; Meng et al. 2015). Till now, mostly RAPD and ISSR markers have been used to analyze genetic relatedness. Recently, Meng et al. (2014) have analyzed the genetic diversity in *Aconitum kongboense* using AFLP markers. Therefore, scope still remains for the use of more robust molecular markers, such as RFLP, AFLP, SSR, SNP, etc for the systematic documentation of fingerprints, efficient germplasm characterization, and identification of superior genotypes.

In conclusion, medicinal plants are the area which requires intensive efforts for which the world demand continues to increase. There is a need for improved techniques of propagation to rapid supply of large number of plants. The synthetic seed technology using alginate encapsulation developed for *Aconitum* species could be useful in germplasm conservation, distribution, and exchange. The ultimate aim of any in vitro propagation system is large-scale multiplication of plantlets which are phenotypically and genetically true-to-type to their mother plant; otherwise, the advantage of tissue culture to get desirable characters of elite/supreme clones will not be achieved. Estimation of genetic fidelity of micropropagated plants is also an important research task. If done properly, micropropagation of selected suitable clones is poised to make spectacular contribution to the growth of pharmaceutical industry. Based on these preliminary findings, hairy root cultures can be another possible alternative source of secondary metabolite production. However, efficiency of secondary metabolite production is sometimes not so desirable. Improved production of secondary metabolites by the over expression of single genes is another important aspect of metabolic engineering. This approach may lead to enhance enzymes involved in metabolism and accumulation of the target products. Molecular biology of *Aconitum* is at primitive stage, and that is why, priority should be given to work based on DNA markers for construction of genetic linkage maps followed by linkage or association studies for traits of interest, such as higher content of phytochemicals etc. These could be further validated through functional genomics and proteomics techniques. This will be helpful to reveal several ways, such as marker-assisted selection as well as identifying specific gene. In essence, biotechnological interventions have opened up new vistas for propagation, conservation, and improvement of this medicinally important genus *Aconitum*.

Author contribution statement JMR is expert in the current biotechnological aspects and has prepared the overall biotechnological part in the manuscript. BR has written the whole manuscript. He also has helped during the plant collection from the field. AB assisted Dr. BR during collection of information about Aconitum from different libraries and database. SY worked as a team member during collection of information about Aconitum and helped to rewrite the molecular biology part in this MS. SM is working in the field of genetic engineering since last few years and has contributed the genetic transformation part in the manuscript. Dr. AC has been a potential part in the process of review. He has collected information from different sources. Being the senior most author, Dr. SNM has developed the concept and frame of the manuscript. The manuscript was polished time to time by Dr. M.

Acknowledgements Authors are thankful to the Director of the Institute for providing the necessary facilities to carry out this work. Sincere acknowledgment goes to the Department of Science and Technology, Govt. of India for the financial support to JMR (WOSA/LS-191/2010) and BR (SB/YS/LS-205/2013), respectively. Authors are highly thankful to the Society for the Conservation of Nature, Rewa, MP, India for the help and guidance provided during the preparation of this review.

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